Differentiation Rules: Power, Product, Quotient, Chain

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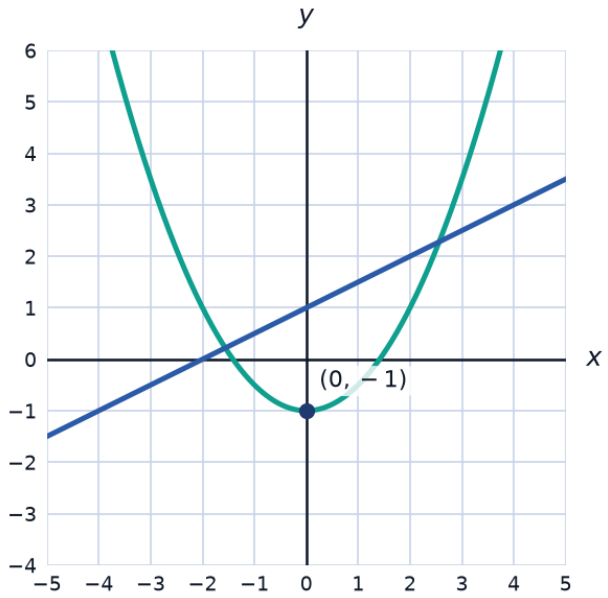
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Power Rule and Basic Operations

Evaluate $\int_2^5 \frac{2x+1}{(x-1)(x+4)}\,dx$.

2 [Foundational] • Target: Product and Quotient Rules

Evaluate $\int_1^5 \frac{1}{(x+1)(x+4)}\,dx$.



3 [Foundational] • Target: The Chain Rule

For parametric $x = e^t \cos t$, $y = e^t \sin t$, find $\frac{d^2y}{dx^2}$ at $t = 0$.

4 [Foundational] • Target: Trigonometric Derivatives with Chain Rule

If $z = x^3 + 3xy^2$ with $x = t^2$ and $y = \sin(t)$, find $\frac{dz}{dt}$ at $t = 1$.

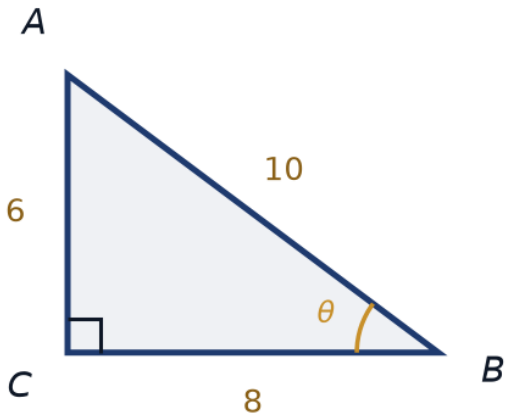
5 [Foundational] • Target: Power Rule and Basic Operations

If $\sin(\theta) = -\frac{24}{25}$ and θ lies in quadrant III, find $\sin 2\theta$.

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6 [Foundational] • Target: Product and Quotient Rules

If $\sin(\theta) = \frac{5}{13}$ and θ lies in quadrant II, find $\cos 2\theta$.



7 [Foundational] • Target: The Chain Rule

Let $G(x) = \int_0^{\sin(x)} \cos(t)\,dt$. Find $G'(-1)$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Trigonometric Derivatives with Chain Rule

Coffee at 70°C is placed in a 20°C room and cools to 60°C in 10 min (Newton's law of cooling). How long (min) until it reaches 40°C?

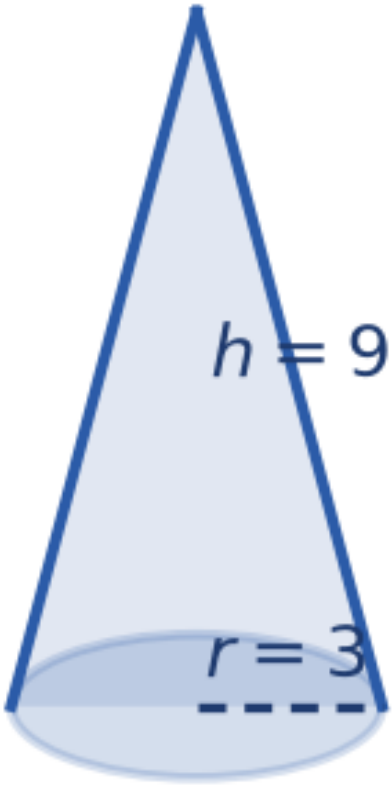
9 [Intermediate] • Target: Power Rule and Basic Operations

Find the sum $\sum_{n=1}^{\infty} \frac{2}{n(n+4)}$.

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10 **[Intermediate]** • **Target: Product and Quotient Rules**

Water pours into an inverted cone (radius 3 m, height 9 m) at 3 m³/min. How fast is the depth rising when it is 8 m? (m/min)



11 **[Intermediate]** • **Target: The Chain Rule**

Product rule for $f = x^2 e^x$:.

12 **[Intermediate]** • **Target: Trigonometric Derivatives with Chain Rule**

Chain rule: $(\sin(e^x))' =$.

13 **[Intermediate]** • **Target: Power Rule and Basic Operations**

Chain rule: $w = f(x, y, z)$, $x = e^s$, $y = e^s \sin t$, $z = e^s \cos t$. $\partial w/\partial s|_{s=0,\, t=0}$:.

14 **[Intermediate]** • **Target: Product and Quotient Rules**

$\frac{d}{dx}[f(g(x))]$ equals (chain rule):.

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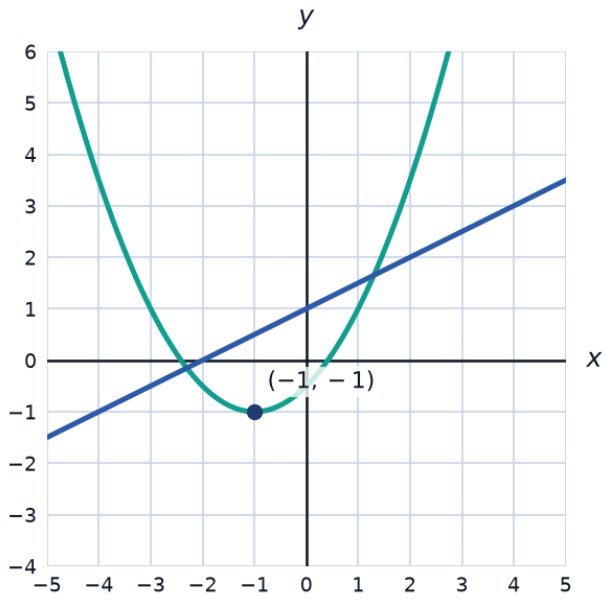
Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: The Chain Rule** • **Cross-Lesson: 12.01 Convergence Tests**

Quotient Rule: $\frac{d}{dx}\left(\frac{x}{x+1}\right) =$.

16 **[Challenging]** • **Target: Trigonometric Derivatives with Chain Rule** • **Cross-Lesson: 12.02 Power Series and Taylor Series**

Using the Product Rule, $\frac{d}{dx}[(3x+1)(x^2-2)] =$.



17 **[Challenging]** • **Target: Power Rule and Basic Operations** • **Cross-Lesson: 13.01 Vectors and Vector-Valued Functions**

$\frac{d}{dx}(x^2-3)^4 =$ (chain rule).

18 **[Challenging]** • **Target: Product and Quotient Rules** • **Cross-Lesson: 14.01 Partial Derivatives and the Gradient**

Product Rule: $\frac{d}{dx}(x\sin x) =$.

19 **[Challenging]** • **Target: The Chain Rule** • **Cross-Lesson: 15.01 Double Integrals**

Chain Rule: $\frac{d}{dx}\sin(x^2) =$.

20 **[Challenging]** • **Target: Trigonometric Derivatives with Chain Rule** • **Cross-Lesson: 16.01 Vector Fields and Line Integrals**

Product of two power series $A(x) = \sum a_n x^n$ and $B(x) = \sum b_n x^n$: coefficient of x^n in $A(x)B(x)$ is:.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Power Rule and Basic Operations

Find the derivative of $f(x) = 4x^5 - 3x^3 + 7x - 9$.

a. $20x^4 - 9x^2 + 7$

b. $20x^4 - 9x^2$

c. $4x^4 - 3x^2 + 7$

d. $20x^4 - 6x^2 + 7$

22 [Foundational] • Target: Power Rule and Basic Operations

Differentiate $g(x) = \frac{4}{\sqrt{x}} + \frac{3}{x^2}$.

a. $-2x^{-3/2} - 6x^{-3}$

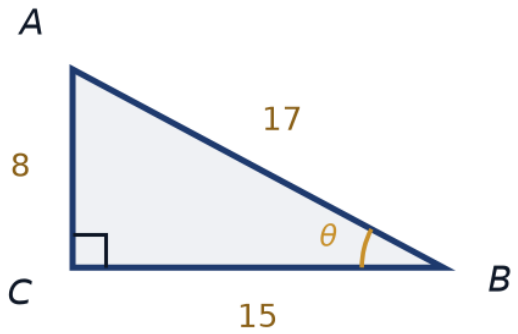
b. $-2x^{-1/2} - 6x^{-3}$

c. $2x^{-3/2} - 6x^{-3}$

d. $-4x^{-3/2} - 3x^{-3}$

23 [Foundational] • Target: Power Rule and Basic Operations

Find all points where the curve $y = x^3 - 3x^2 - 9x + 5$ has a horizontal tangent.



a. $x = -1$ and $x = 3$

b. $x = 1$ and $x = -3$

c. $x = 0$ and $x = 3$

d. $x = -1$ only

24 [Foundational] • Target: Product and Quotient Rules

Differentiate $h(x) = (x^2 + 3)(2x - 5)$.

a. $6x^2 - 10x + 6$

b. $6x^2 - 10x$

c. $4x^2 - 10x + 6$

d. $6x^2 + 6$

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25 [Foundational] • Target: Product and Quotient Rules

Find $\frac{dy}{dx}$ for $y = \frac{3x-1}{2x+5}$.

a. $\frac{17}{(2x+5)^2}$

b. $\frac{1}{(2x+5)^2}$

c. $\frac{6x+17}{(2x+5)^2}$

d. $\frac{3}{2}$

26 [Foundational] • Target: Product and Quotient Rules

If $f(2) = 3, f'(2) = -1, g(2) = 4, g'(2) = 2$, find $\left(\frac{f}{g}\right)'(2)$.

a. $-10/16 = -5/8$

b. $2/16 = 1/8$

c. $-2/16$

d. $-1/2$

27 [Foundational] • Target: The Chain Rule

Find $\frac{d}{dx}[(3x^2 - 5x + 2)^4]$.

a. $4(3x^2 - 5x + 2)^3(6x - 5)$

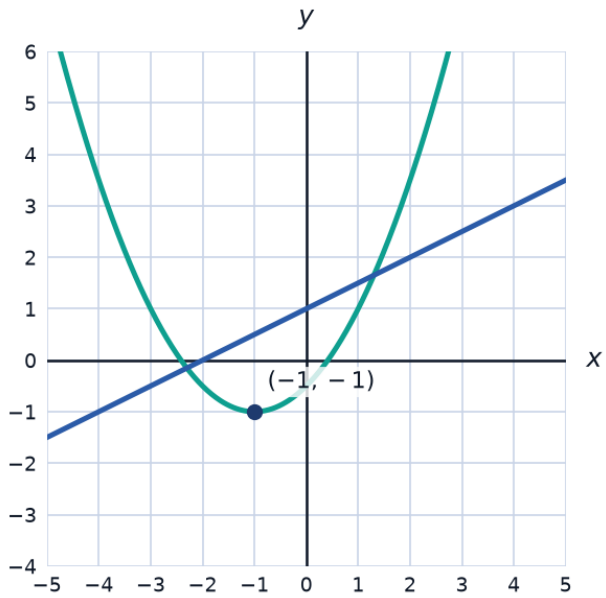
b. $4(3x^2 - 5x + 2)^3$

c. $12(6x - 5)^3$

d. $4(6x - 5)^3$

28 [Foundational] • Target: The Chain Rule

Find $\frac{dy}{dx}$ for $y = \sqrt{4 - x^2}$.



a. $\frac{-x}{\sqrt{4-x^2}}$

b. $\frac{x}{\sqrt{4-x^2}}$

c. $\frac{-1}{2\sqrt{4-x^2}}$

d. $\frac{-2x}{\sqrt{4-x^2}}$

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Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 **[Intermediate]** • **Target: The Chain Rule**

If $h(x) = f(g(x))$ with $g(3) = 5$, $g'(3) = 4$, and $f'(5) = -2$, find $h'(3)$.

a. -8

b. 8

c. -10

d. 20

30 **[Intermediate]** • **Target: Trigonometric Derivatives with Chain Rule**

Differentiate $y = \sin(3x^2)$.

a. $6x\cos(3x^2)$

b. $3\cos(3x^2)$

c. $\cos(6x)$

d. $6x\sin(3x^2)$

31 **[Intermediate]** • **Target: Trigonometric Derivatives with Chain Rule**

Find $\frac{d}{dx}[\tan^3(x)]$.

a. $3\tan^2(x)\sec^2(x)$

b. $3\tan^2(x)$

c. $3\sec^4(x)$

d. $\sec^6(x)$

32 **[Intermediate]** • **Target: Trigonometric Derivatives with Chain Rule**

Find the second derivative y'' for $y = \cos(2x)$.

a. $-4\cos(2x)$

b. $4\cos(2x)$

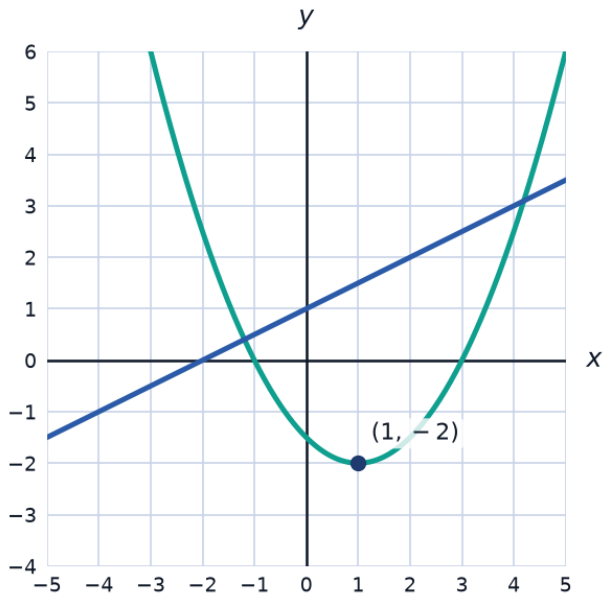
c. $-2\sin(2x)$

d. $-4\sin(2x)$

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33 **[Intermediate]** • **Target: Power Rule and Basic Operations**

Implicit differentiation: $\frac{d^2y}{dx^2}$ at $(1, 1)$ for $x^2 + y^2 = 2$:



a. -1

b. -2

c. 0

d. 1

34 **[Intermediate]** • **Target: Product and Quotient Rules**

Cauchy product: $(\sum_{n=0}^{\infty} x^n)^2 = \sum_{n=0}^{\infty} (n+1)x^n$. This equals:

a. $1/(1-x)^2$

b. $1/(1-x)$

c. $(1-x)^2$

d. $2/(1-x)$

35 **[Intermediate]** • **Target: The Chain Rule**

Implicit differentiation of $F(x, y) = 0$: $\frac{dy}{dx} =$

a. $-F_x/F_y$

b. F_x/F_y

c. $-F_y/F_x$

d. F_{xy}

36 **[Intermediate]** • **Target: Trigonometric Derivatives with Chain Rule**

Implicit differentiation of $x^2 + y^2 = 25$: $dy/dx =$

a. $-x/y$

b. x/y

c. $-y/x$

d. $2x$

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37 **[Intermediate]** • **Target: Power Rule and Basic Operations**

Term-by-term differentiation is valid inside the radius of convergence. The derivative of $\sum_{n=0}^{\infty} x^n/(n+1)$ is:

a. $\sum x^n/n$

b. $\sum n x^{n-1}/(n+1)$

c. $1/(1-x)$

d. $\ln(1-x)$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 **[Challenging]** • **Target: Product and Quotient Rules** • **Cross-Lesson: 14.01 Partial Derivatives and the Gradient**

Logarithmic differentiation: if $y = \frac{x^2\sqrt{x+1}}{e^x}$, then $y' =$

a. $y\left(\frac{2}{x} + \frac{1}{2(x+1)} - 1\right)$

b. $y\left(\frac{2}{x} - \frac{1}{2(x+1)} - 1\right)$

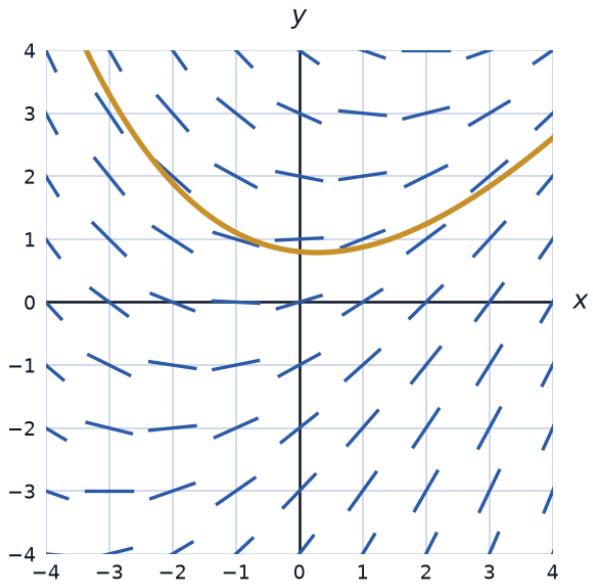
c. $y\left(\frac{2}{x} + \frac{1}{2(x+1)} + 1\right)$

d. $y\left(2x + \frac{1}{2} - 1\right)$

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39 [Challenging] • Target: The Chain Rule • Cross-Lesson: 15.01 Double Integrals

Logarithmic differentiation: $y = x^x$. $y' =$



a. $x^x(1 + \ln x)$

b. $x \cdot x^{x-1}$

c. $x^x \ln x$

d. e^x

40 [Challenging] • Target: Trigonometric Derivatives with Chain Rule • Cross-Lesson: 16.01 Vector Fields and Line Integrals

The power series $\sum a_n x^n$ represents a function that is C^∞ (infinitely differentiable) inside its radius of convergence. This means all derivatives:

a. are polynomials

b. are zero

c. exist

d. diverge

41 [Challenging] • Target: Power Rule and Basic Operations • Cross-Lesson: 2.01 Evaluating Limits

Differentiation of geometric series $1/(1-x) = \sum x^n$ gives $1/(1-x)^2 =$

a. $\sum n x^{n-1}$

b. $\sum (n+1) x^n$

c. $\sum x^{n-1}$

d. $\sum n x^n$

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42 [Challenging] • Target: Product and Quotient Rules • Cross-Lesson: 2.02 Continuity

Implicit differentiation of $xy = 5$: $\frac{dy}{dx} =$

a. $\frac{5}{x}$

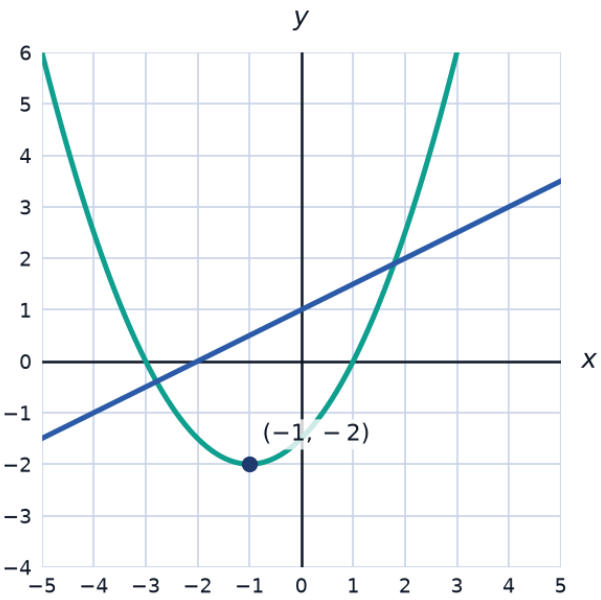
b. $\frac{y}{x}$

c. $-\frac{y}{x}$

d. $\frac{x}{y}$

43 [Challenging] • Target: The Chain Rule • Cross-Lesson: 3.01 The Derivative from First Principles

Leibniz rule: $(fg)^{(n)} = \sum_{k=0}^{n} \binom{n}{k} f^{(k)} g^{(n-k)}$. Analogy with:



a. binomial theorem

b. FTC

c. MVT

d. Taylor series

44 [Challenging] • Target: Trigonometric Derivatives with Chain Rule • Cross-Lesson: 4.01 Antiderivatives and Indefinite Integrals

Term-by-term differentiation of $\sum a_n x^n$ within $|x| < R$ gives:

a. $\sum n a_n x^n$

b. $\sum a_{n+1} x^{n+1}$

c. $\sum a_n x^{n+1}/(n+1)$

d. $\sum n a_n x^{n-1}$

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45 [Challenging] • Target: Power Rule and Basic Operations • Cross-Lesson: 4.02 Riemann Sums and Definite Integrals

L'Hôpital's rule applies to $\lim_{x \to a} \frac{f}{g}$ when the limit is of the form:

a. 0/1

b. 1/0

c. 0/0

d. 1/1

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